```python
DNANucleotides = ['A', 'C','G','T']
dna_seq = input("Type your DNA Strand here:") # You input your DNA strand

# Function 1: Validating a sequence if its true ------------------------------------
def ValidateSeq(dna_seq):
  tmpseq = dna_seq.upper()
  for nuc in tmpseq:
      if nuc not in DNANucleotides:
        return print("Sorry that's not a DNA Strand :(")
        break
  return tmpseq
print("Yay, this is your DNA Strand:" + ValidateSeq(dna_seq)) #


# Function 2 Random DNA Generator --------------------------------------------------
import random
randDNAstr = ''.join([random.choice(DNANucleotides) for nuc in range(20)])
print("Here is a randomly generated DNA: " + ValidateSeq(randDNAstr))
DNAstring = ValidateSeq(randDNAstr)

# Function 3 Counting Nucleotides ---------------------------------------------------
def countNucfrequency(seq):
  temporaryFrequencyDictionary = {"A": 0, "C": 0, "G": 0, "T": 0}
  for nuc in DNAstring:
    temporaryFrequencyDictionary[nuc] += 1
  return temporaryFrequencyDictionary
print("DNA Strand:" + ValidateSeq(randDNAstr))
print(countNucfrequency(DNAstring))


# Function 4 Transcription DNA -> RNA ------------------------------------------------------
def transcription(seq):
  return seq.replace("T", "U")
print("DNA Strand:" + ValidateSeq(randDNAstr)) # Makes sure we use DNA
print(countNucfrequency(DNAstring)) # Counts Bases
print("RNA Strand" + transcription(DNAstring)) # Turned DNA -> RNA
```

```
Type your DNA Strand here:CCGAC
Yay, this is your DNA Strand:CCGAC
Here is a randomly generated DNA: TGTTAGTGGCAGCCGCTTAG
DNA Strand:TGTTAGTGGCAGCCGCTTAG
{'A': 3, 'C': 4, 'G': 7, 'T': 6}
DNA Strand:TGTTAGTGGCAGCCGCTTAG
{'A': 3, 'C': 4, 'G': 7, 'T': 6}
RNA StrandUGUUAGUGGCAGCCGCUUAG
```